S4-GEO Stage 3

Equality Extension — bar geometry vs next-bar direction, tick-exact

tf	data span (UTC)	rows	recovered tie-bars	baseline UP	flags
1m	Jun 28 11:03 -> Jul 02 11:17	9192	6946 (75.6%)	50.34%	SPENT
5m	Jun 21 06:00 -> Jul 02 11:15	5372	2688 (50.0%)	50.13%	
15m	Jun 21 06:00 -> Jul 02 11:14	1818	590 (32.5%)	48.40%	
1h	Jun 21 06:49 -> Jul 02 11:17	457	82 (17.9%)	47.48%	
4h	Jun 21 06:35 -> Jul 02 11:14	112	12 (10.7%)	46.43%	THIN

Tick: $0.01, derived from data — every O/C/H/L and ladder price on all five timeframes is cent-aligned (gcd of scaled prices = 1 cent). Equality means equal after rounding to this tick.

Honesty framing. Tie-bar cells (the “=” pairs, multi-equalities, and the M=P-collapsed orderings) are the FIRST ANALYSIS of bars every earlier stage excluded. Extended non-tie cells (strict pairs, or-tied extremes) are re-cuts of already-mined data and are characterization, not discovery. The 1m dataset is spent throughout. Outcome is next-bar direction — an information measure: no barriers, no fees, not profitability. Holdouts (last 30%, 1-bucket embargo) were sealed and judged once.

Multiplicity. This stage screens 40 cells × 5 timeframes = 200. S4-GEO running total: 50 (stage 1) + 200 (stage 2) + 200 (stage 3) = **450 cells**, on top of ~35 earlier trials against the spent 1m data.

1m — SPENT (characterization only)

*Cells leaning past 50% next-bar direction, sorted by |lift|. Baseline P(UP) = **50.34%** over 9192 rows; survivor min-n 100; grey = below min-n. * = sealed-holdout PASS.*

shape	n	P(UP)	P(DOWN)	lift	holdout
MP>C>O (M=P bars)	167	**56.3%**	43.7%	+6.0 pp	-
MP>O>C (M=P bars)	198	44.4%	**55.6%**	-5.9 pp	-
C>MP>O (M=P bars)	911	**55.8%**	44.2%	+5.4 pp	**PASS ***
O lowest-or-tied	4022	**53.1%**	46.9%	+2.8 pp	-
C lowest-or-tied	4055	47.5%	**52.5%**	-2.8 pp	-
C > M	3932	**53.0%**	47.0%	+2.6 pp	-
C highest-or-tied	4097	**52.9%**	47.1%	+2.6 pp	-
O < C	4290	**52.8%**	47.2%	+2.5 pp	-
C>O>MP (M=P bars)	193	**52.9%**	47.1%	+2.5 pp	-
O > C	4274	47.9%	**52.1%**	-2.5 pp	-
C < M	3880	48.1%	**51.9%**	-2.2 pp	-
O < M	3859	**52.5%**	47.5%	+2.1 pp	-
C < P	3662	48.2%	**51.8%**	-2.1 pp	-
O < P	3690	**52.4%**	47.6%	+2.1 pp	-
O=M & C=P	271	**52.4%**	47.6%	+2.1 pp	-
O highest-or-tied	4027	48.3%	**51.7%**	-2.0 pp	-
O > M	3869	48.4%	**51.6%**	-2.0 pp	-
C > P	3759	**52.3%**	47.7%	+2.0 pp	-
O > P	3732	48.7%	**51.3%**	-1.7 pp	-
C=M=P	937	48.7%	**51.3%**	-1.7 pp	-
O>C>MP (M=P bars)	192	49.0%	**51.0%**	-1.4 pp	-
C = M	1380	49.0%	**51.0%**	-1.4 pp	-
O=C=M	217	**51.6%**	48.4%	+1.3 pp	-
O=C & M=P	380	**51.6%**	48.4%	+1.2 pp	-
P lowest-or-tied	2948	49.3%	**50.7%**	-1.1 pp	-
O>MP>C (M=P bars)	929	49.3%	**50.7%**	-1.0 pp	-
M < P	2274	49.5%	**50.5%**	-0.9 pp	-
O = P	1770	49.5%	**50.5%**	-0.8 pp	-
M = P	4518	**51.0%**	49.0%	+0.6 pp	-
O=C=M=P	175	**50.9%**	49.1%	+0.5 pp	-
O=P & C=M	248	**50.8%**	49.2%	+0.5 pp	-
P highest-or-tied	2868	49.9%	**50.1%**	-0.4 pp	-
C = P	1771	**50.5%**	49.5%	+0.2 pp	-
O = C (doji bar)	628	**50.2%**	49.8%	-0.2 pp	-
O=C=P	253	**50.2%**	49.8%	-0.1 pp	-
M highest-or-tied	1869	**50.4%**	49.6%	+0.1 pp	-
O=M=P	961	**50.4%**	49.6%	+0.0 pp	-
M lowest-or-tied	1819	**50.4%**	49.6%	+0.0 pp	-

5m

*Cells leaning past 50% next-bar direction, sorted by |lift|. Baseline P(UP) = **50.13%** over 5372 rows; survivor min-n 100; grey = below min-n. * = sealed-holdout PASS.*

shape	n	P(UP)	P(DOWN)	lift	holdout
C>O>MP (M=P bars)	80	60.0%	40.0%	+9.9 pp	-
O=C=M=P	27	59.3%	40.7%	+9.1 pp	-
O=C=M	38	57.9%	42.1%	+7.8 pp	-
MP>O>C (M=P bars)	104	43.3%	**56.7%**	-6.9 pp	-
O=M & C=P	45	55.6%	44.4%	+5.4 pp	-
O=P & C=M	54	55.6%	44.4%	+5.4 pp	-
O>MP>C (M=P bars)	361	44.9%	**55.1%**	-5.2 pp	-
O=C=P	47	55.3%	44.7%	+5.2 pp	-
O=M=P	186	**54.8%**	45.2%	+4.7 pp	-
MP>C>O (M=P bars)	87	54.0%	46.0%	+3.9 pp	-
O = C (doji bar)	175	**53.7%**	46.3%	+3.6 pp	-
O = P	567	**53.4%**	46.6%	+3.3 pp	-
O=C & M=P	79	46.8%	53.2%	-3.3 pp	-
O>C>MP (M=P bars)	98	46.9%	53.1%	-3.2 pp	-
C highest-or-tied	2202	**52.9%**	47.1%	+2.7 pp	-
O highest-or-tied	2188	47.8%	**52.2%**	-2.4 pp	-
O lowest-or-tied	2177	**52.3%**	47.7%	+2.2 pp	-
O > C	2586	48.0%	**52.0%**	-2.2 pp	-
C > P	2411	**52.1%**	47.9%	+2.0 pp	-
O < C	2611	**52.0%**	48.0%	+1.9 pp	-
C < M	2480	48.4%	**51.6%**	-1.8 pp	-
C>MP>O (M=P bars)	380	**51.8%**	48.2%	+1.7 pp	fail
O > P	2416	48.5%	**51.5%**	-1.7 pp	-
C > M	2493	**51.8%**	48.2%	+1.6 pp	-
O > M	2498	48.5%	**51.5%**	-1.6 pp	-
C < P	2413	48.5%	**51.5%**	-1.6 pp	-
C = P	548	48.5%	**51.5%**	-1.6 pp	-
O < M	2464	**51.5%**	48.5%	+1.4 pp	-
C lowest-or-tied	2185	48.7%	**51.3%**	-1.4 pp	-
O = M	410	**51.5%**	48.5%	+1.3 pp	-
P highest-or-tied	1351	48.9%	**51.1%**	-1.3 pp	-
M highest-or-tied	710	**51.3%**	48.7%	+1.1 pp	-
C=M=P	174	**51.1%**	48.9%	+1.0 pp	-
O < P	2389	**51.0%**	49.0%	+0.9 pp	-
C = M	399	**50.9%**	49.1%	+0.8 pp	-
P lowest-or-tied	1367	**50.8%**	49.2%	+0.7 pp	-
M lowest-or-tied	689	49.5%	**50.5%**	-0.6 pp	-
M = P	1495	49.6%	**50.4%**	-0.6 pp	-
M > P	1949	**50.5%**	49.5%	+0.4 pp	-

shape	n	P(UP)	P(DOWN)	lift	holdout
M < P	1928	**50.2%**	49.8%	+0.0 pp	-

15m

*Cells leaning past 50% next-bar direction, sorted by |lift|. Baseline P(UP) = **48.40%** over 1818 rows; survivor min-n 50; grey = below min-n. * = sealed-holdout PASS.*

shape	n	P(UP)	P(DOWN)	lift	holdout
O=P & C=M	3	100.0%	0.0%	+51.6 pp	-
O=C=M=P	2	100.0%	0.0%	+51.6 pp	-
O=M & C=P	6	83.3%	16.7%	+34.9 pp	-
O=C=P	5	80.0%	20.0%	+31.6 pp	-
O=C & M=P	10	80.0%	20.0%	+31.6 pp	-
C>O>MP (M=P bars)	29	79.3%	20.7%	+30.9 pp	-
O=C=M	4	75.0%	25.0%	+26.6 pp	-
O>C>MP (M=P bars)	29	69.0%	31.0%	+20.6 pp	-
MP>O>C (M=P bars)	17	64.7%	35.3%	+16.3 pp	-
M lowest-or-tied	202	**63.9%**	36.1%	+15.5 pp	**PASS ***
C=M=P	23	60.9%	39.1%	+12.5 pp	-
O = C (doji bar)	32	59.4%	40.6%	+11.0 pp	-
O=M=P	21	57.1%	42.9%	+8.7 pp	-
O>MP>C (M=P bars)	84	41.7%	**58.3%**	-6.7 pp	-
O = M	71	**54.9%**	45.1%	+6.5 pp	-
MP>C>O (M=P bars)	22	54.5%	45.5%	+6.1 pp	-
M = P	307	**54.4%**	45.6%	+6.0 pp	fail
P lowest-or-tied	445	**53.3%**	46.7%	+4.8 pp	fail
C < M	877	44.0%	**56.0%**	-4.4 pp	-
C > M	862	**52.8%**	47.2%	+4.4 pp	-
C lowest-or-tied	715	44.1%	**55.9%**	-4.3 pp	-
C = P	105	**51.4%**	48.6%	+3.0 pp	-
C < P	863	45.4%	**54.6%**	-3.0 pp	-
C > P	850	**51.1%**	48.9%	+2.6 pp	-
O highest-or-tied	709	46.1%	**53.9%**	-2.3 pp	-
O > C	926	46.4%	**53.6%**	-2.0 pp	-
C highest-or-tied	672	**50.1%**	49.9%	+1.7 pp	-
O < C	860	**50.1%**	49.9%	+1.7 pp	-
P highest-or-tied	431	**50.1%**	49.9%	+1.7 pp	-
O = P	109	46.8%	**53.2%**	-1.6 pp	-
M < P	768	47.0%	**53.0%**	-1.4 pp	-
O lowest-or-tied	666	49.5%	**50.5%**	+1.1 pp	-
C>MP>O (M=P bars)	76	47.4%	**52.6%**	-1.0 pp	-
M > P	743	47.4%	**52.6%**	-1.0 pp	-
C = M	79	49.4%	**50.6%**	+1.0 pp	-
O < P	835	49.2%	**50.8%**	+0.8 pp	-
M highest-or-tied	175	49.1%	**50.9%**	+0.7 pp	-
O > M	887	47.7%	**52.3%**	-0.7 pp	-
O > P	874	47.8%	**52.2%**	-0.6 pp	-

shape	n	P(UP)	P(DOWN)	lift	holdout
O < M	860	48.6%	**51.4%**	+0.2 pp	-

1h

*Cells leaning past 50% next-bar direction, sorted by |lift|. Baseline P(UP) = **47.48%** over 457 rows; survivor min-n 25; grey = below min-n. * = sealed-holdout PASS.*

shape	n	P(UP)	P(DOWN)	lift	holdout
O=C=P	1	100.0%	0.0%	+52.5 pp	-
O=M=P	2	100.0%	0.0%	+52.5 pp	-
C=M=P	1	100.0%	0.0%	+52.5 pp	-
O>C>MP (M=P bars)	3	100.0%	0.0%	+52.5 pp	-
O=M & C=P	1	0.0%	100.0%	-47.5 pp	-
MP>O>C (M=P bars)	7	71.4%	28.6%	+23.9 pp	-
O = C (doji bar)	3	66.7%	33.3%	+19.2 pp	-
MP>C>O (M=P bars)	3	66.7%	33.3%	+19.2 pp	-
C = M	12	58.3%	41.7%	+10.8 pp	-
M = P	36	**58.3%**	41.7%	+10.8 pp	fail
C>MP>O (M=P bars)	8	37.5%	62.5%	-10.0 pp	-
C = P	14	57.1%	42.9%	+9.7 pp	-
M lowest-or-tied	41	**56.1%**	43.9%	+8.6 pp	-
M highest-or-tied	40	**55.0%**	45.0%	+7.5 pp	fail
O>MP>C (M=P bars)	10	40.0%	60.0%	-7.5 pp	-
O = M	13	53.9%	46.1%	+6.4 pp	-
O = P	13	53.9%	46.1%	+6.4 pp	-
C highest-or-tied	161	**52.8%**	47.2%	+5.3 pp	**PASS ***
O highest-or-tied	180	42.2%	**57.8%**	-5.3 pp	-
O lowest-or-tied	167	**52.7%**	47.3%	+5.2 pp	-
O < C	217	**52.1%**	47.9%	+4.6 pp	-
O > C	237	43.0%	**57.0%**	-4.5 pp	-
M > P	208	43.3%	**56.7%**	-4.2 pp	-
C < M	225	43.6%	**56.4%**	-3.9 pp	-
O > M	229	43.7%	**56.3%**	-3.8 pp	-
O < M	215	**51.2%**	48.8%	+3.7 pp	-
C lowest-or-tied	168	44.0%	**56.0%**	-3.4 pp	-
C > M	220	**50.9%**	49.1%	+3.4 pp	-
O > P	225	44.4%	**55.6%**	-3.0 pp	-
O < P	219	**50.2%**	49.8%	+2.7 pp	-
M < P	213	49.8%	**50.2%**	+2.3 pp	-
C < P	223	46.2%	**53.8%**	-1.3 pp	-
P lowest-or-tied	106	46.2%	**53.8%**	-1.3 pp	-
C > P	220	48.2%	**51.8%**	+0.7 pp	-
P highest-or-tied	105	47.6%	**52.4%**	+0.1 pp	-

4h — THIN (114 usable bars; all rows greyed)

*Cells leaning past 50% next-bar direction, sorted by |lift|. Baseline P(UP) = **46.43%** over 112 rows; survivor min-n 14; grey = below min-n or THIN tf. * = sealed-holdout PASS.*

shape	n	P(UP)	P(DOWN)	lift	holdout
O = C (doji bar)	1	100.0%	0.0%	+53.6 pp	-
C>O>MP (M=P bars)	1	100.0%	0.0%	+53.6 pp	-
MP>C>O (M=P bars)	2	100.0%	0.0%	+53.6 pp	-
O = M	1	0.0%	100.0%	-46.4 pp	-
O = P	1	0.0%	100.0%	-46.4 pp	-
C>MP>O (M=P bars)	1	0.0%	100.0%	-46.4 pp	-
O>MP>C (M=P bars)	1	0.0%	100.0%	-46.4 pp	-
P highest-or-tied	24	70.8%	29.2%	+24.4 pp	fail
P lowest-or-tied	31	32.3%	67.7%	-14.2 pp	fail
M = P	5	60.0%	40.0%	+13.6 pp	-
M lowest-or-tied	10	60.0%	40.0%	+13.6 pp	-
C < P	50	58.0%	42.0%	+11.6 pp	fail
M highest-or-tied	7	57.1%	42.9%	+10.7 pp	-
C > P	60	36.7%	63.3%	-9.8 pp	fail
O < P	51	54.9%	45.1%	+8.5 pp	-
C < M	51	54.9%	45.1%	+8.5 pp	-
C > M	59	39.0%	61.0%	-7.5 pp	PASS *
O highest-or-tied	46	39.1%	60.9%	-7.3 pp	-
O > P	60	40.0%	60.0%	-6.4 pp	-
M > P	57	40.4%	59.6%	-6.1 pp	fail
C highest-or-tied	37	40.5%	59.5%	-5.9 pp	-
M < P	50	52.0%	48.0%	+5.6 pp	fail
O < M	53	50.9%	49.1%	+4.5 pp	-
O > M	58	43.1%	56.9%	-3.3 pp	-
O > C	60	45.0%	55.0%	-1.4 pp	-
O < C	51	47.1%	52.9%	+0.6 pp	-

Cross-timeframe summary — holdout PASSes only

tf	shape	class	disc n / lift	90% CI	hold n / lift
1m	C>MP>O (M=P bars)	EQ	616 / 5.71 pp	[1.68, 9.74]	295 / 4.75 pp
15m	M lowest-or-tied	RECUT	134 / 16.12 pp	[8.81, 24.22]	67 / 13.37 pp
1h	C highest-or-tied	RECUT	113 / 5.5 pp	[0.12, 11.33]	47 / 5.68 pp
4h	C > M	RECUT	38 / -9.45 pp	[-18.07, -0.4]	20 / -5.61 pp

Caveats

• The 1m slice is spent (~35+ cumulative trials plus stages 1–3); its PASS is a characterization, not a confirmed edge — and its 5m sibling cell flipped sign in holdout.

• The HTF window is ~11 days (June 21 – July 2). Day-block bootstrap runs on 4–12 blocks; regime coverage is one market phase. 4h is THIN (114 bars) throughout.

• Equality is tick-exact at $0.01. A different binning (e.g. coarser “near-equal” bands) was NOT tested and would be a new multiplicity charge.

• Non-tie cells re-cut data already mined by stages 1–2 on the same window; only the tie-bar cells are first-look. All holdouts were judged exactly once — these verdicts are final for this snapshot; next evidence must come from forward snapshots.